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G H Raisoni International Skill Tech University, Pune

S.Y. B.Tech (Branch Name)

(Sem-IV)

CAE II- SUMMER 2026 (20_ Pattern)

Operating Systems (25BTPCL403)

(Time: - 01 Hour) (Max. Marks: 20)

Instructions to the students:

i) Solve Q.1 or 2 and Q3 or 4. ii) Neat diagrams must be drawn wherever necessary.
iii) Figures to the right indicate full marks. iv) Use of scientific calculator is allowed
v) Assume suitable data, if necessary.

Q.No	Question	Marks	CO	BL
Q1.(a)	Explain the concept of Process Synchronization	[5]	III	2
b)	Explain in detail concept of Semaphore	[5]	III	2
	OR			
Q 2.(a)	Describe the concept of monitors and components of monitors	[5]	III	2
(b)	Differentiate Monitor and Semaphore	[5]	III	2
Q 3.(a)	Illustrate deadlock detection	[5]	IV	3
(b)	Illustrate deadlock recovery	[5]	IV	3
	OR			
Q. 4 (a)	What is the Resource Allocation Graph	[5]	IV	2
(b)	Explain Coffman conditions	[5]	IV	2

- **BL – Bloom's Taxonomy Levels (1- Remember, 2- Understand, 3 – Apply, 4 – Analyze,5 –Evaluate 6 - Create)**
- **CO- Course Outcomes**